Review

Pharmacovigilance: An Overview

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ABSTRACT

Purpose: Pharmacovigilance (PV) is a relatively new discipline in the pharmaceutical industry. Having undergone rapid growth over the past 2 decades, PV now touches many other disciplines in the research and development enterprise. With its growth has come a heightened awareness and interest in the medical community about the roles that PV plays. This article provides insights into the background and inner workings of PV.

Methods: This narrative review covers the core PV activities and other major areas of the pharmaceutical enterprise in which PV makes significant contributions.

Findings: Drug safety monitoring activities were organized by the US Food and Drug Administration and academic medical centers in the early 1950s in response to growing concern over the occurrence of aplastic anemia and other blood dyscrasias associated with the use of chloramphenicol. This experience was codified in the 1962 Kefauver-Harris Amendments to the Federal Food, Drug and Cosmetic Act as adverse event evaluation and reporting requirements. The ensuing decades have seen the development of core PV functions for pharmaceutical companies: case management, signal management, and benefit-risk management. A broader scope of PV has developed to include the following major activities: support of patient safety during the conduct of clinical trials through assuring proper use of informed consent and institutional review boards (ethics committees); selection of the first safe dose for use in humans, based on pharmacologic data obtained in animal studies; development of the safety profile for proper use of a new molecular entity and appropriate communication of that information to the range of relevant stakeholders; attendance to surveillance activities through a set of signal management processes; monitoring the manufactured product itself through collaborative activities with manufacturing professionals; management of benefit–risk to assure appropriate use in medical care after marketing; and maintenance of inspection readiness as a corporate cultural process.

Implications: The extent and pace of change promise to accelerate with the integration of biomedical informatics, analytics, artificial intelligence, and machine learning. This progress has implications for the development of the next generation of PV professionals who will need to be trained in entirely new skill sets to lead continued improvements in the safe use of pharmaceuticals. (*Clin Ther*. 2018;40:1991–2004) © 2018 Elsevier Inc. All rights reserved.

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INTRODUCTION

A century-long history of many tragic events has played a critical role in shaping the present-day drug development structures and processes, none more so than those concerned with pharmacovigilance (PV).[1,2] The present review describes the core PV functions of case management, signal management, and benefit–risk management. It also covers the breadth of scope of safety-related activities that a present-day pharmaceutical company must be prepared to manage, most of which are likely to reside in a department charged with PV responsibilities. This review does not concern safety-related issues of

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medical devices, the intricacies of combination products, or companion diagnostics.

The World Health Organization has defined PV as the "science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other possible drug-related problems."[3] PV is commonly considered to have begun as a separate, identifiable activity in the United States with passage of the Drug Efficacy Amendments (also known as the Kefauver-Harris Amendments) of 1962, that were legislated in response to the thalidomide catastrophe that had occurred in Europe.[4] However, there is an earlier history that played an important role in shaping the statute. As early as 1952, the US Food and Drug Administration (FDA) had actively sought information concerning a growing safety issue with the use of chloramphenicol. FDA conducted a survey through its 16 district offices of hospitals, clinics, and medical schools in cities with populations >100,000 to collect reports of aplastic anemia and blood dyscrasias associated with use of chloramphenicol.[5] Thus, a pilot activity of careful, systematic review of individual case reports organized in response to a drug safety concern was codified in the Kefauver-Harris Amendments of 1962 as a required corporate responsibility of evaluation and timely reporting. This early response has continued to evolve into the present-day core function of Case Management.

The task of standardizing definitions and processes was taken up by the Council for International Organization of Medical Sciences (CIOMS).[6] Initially established in 1949 as a nongovernmental organization for international exchange of knowledge in the medical sciences, CIOMS was reorganized in 1966 to include activities such as harmonization of infrastructural concepts, definitions, and processes for the safe use of pharmaceutical products, including medicines and vaccines, through contributions from stakeholders with subject matter expertise. CIOMS used a global working group mechanism of collaborative subject matter experts drawn from government, academia, and industry to tackle newly developing issues in this arena; some of these issues include adverse drug reactions, periodic reporting, risk–benefit, safety reporting in clinical trials, signal detection, and risk minimization.

Another important, complementary organization is the International Council for Harmonisation (ICH) (formerly the International Conference for Harmonisation).[7] Organized in 1990, the ICH has provided a forum for the regulatory agencies and pharmaceutical industries of Europe, Japan, and the United States to harmonize the regulatory infrastructures of these regions. It has since expanded its international outreach and inclusion of stakeholders.

The term "pharmacovigilance," first proposed in the 1970s, has gradually gained traction to become one of the two common terms of art for the overall discipline,[8] the other, older term being "drug safety."

Systematic review of postmarketing individual case study reports also included reports submitted by clinical investigators from earlier in the life-cycle of a new molecular entity (NME), also known as a new chemical entity, that is under development. Gradually, individual reviews were accompanied by aggregate review methods, directed by regulation and guidance documents.[9,10]

Once there were processes for providing standardized adverse event information and for entering it into a database, the next critical step was to develop methods to query the database to determine whether there were adverse events being reported in association with the use of the pharmaceutical agent of interest. Soon thereafter, standardized signal detection activities followed, including data mining.[11,12] CIOMS eventually introduced the comprehensive term "signal management" in 2010,[13] the second core PV function.

Also in the early 2000s, the basis of regulatory assessment expanded to include "risk," and approaches to creating risk management measures were instituted: in the United States, first as risk minimization action plans, then as risk evaluation and mitigation strategy (REMS),[14] and in the European Union, as risk management plans.[15] This measure is the third core PV function: benefit–risk management.

Thus, 3 core functions of PV exist: case management, signal management, and benefit–risk management. To understand how these activities are related to one another, however, it is helpful to look at them from a systems perspective (Figure 1).

A basic, open organizational system consists of 3 components: input, process, and output.[16] Regarding a PV system, the database is the central component, "process," which is essential to all of the core functions. Beginning with the database in the

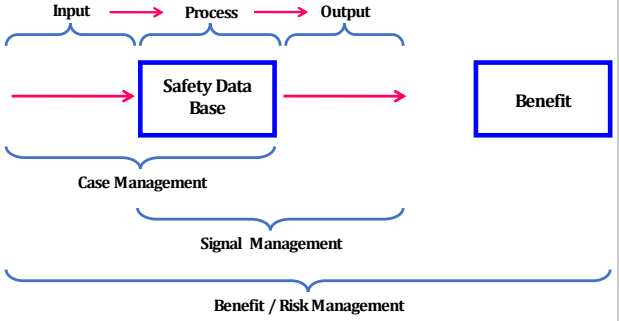


Figure 1. Pharmacovigilance: a systems perspective.

Figure 1, it is the central repository of all patient safety-related information concerning a company's pharmaceutical products; that is, its drugs, biologics, vaccines, medical devices, combination products, and in vitro diagnostics. Ideally, this information identifies a reporter and a patient (eg, demographic characteristics, concurrent illnesses), specifies the pharmaceutical agent of concern (eg, dosage, route of administration, duration, other concurrent medications), and details about the adverse event that was experienced (eg, duration, complications, hospitalization). Whatever the source of the information that a company becomes aware of, regulations stipulate the systematic and standardized preparation of the information for timely reporting.[17] During the early years of PV activities, the database was a modest, manually organized spreadsheet; with time, it gradually developed into an electronic, software spreadsheet, and eventually it took on its present-day form as a commercial, highly functional, dedicated database.[18,19] In addition to individual companies maintaining such databases, regulatory agencies also maintain massive databases of adverse event reports from companies that have products approved within their jurisdictions, routinely receiving >1 million reports per year.

Moving to "input" in Figure 1, the safety physician in those early years was primarily focused on learning how to take information from CIOMS adverse event reporting forms[20] (and later to include the FDA MedWatch forms[21]) and put it into the database in a consistent, standardized, and timely way to meet compliance requirements. This goal necessitated a globally standardized coding system that has come to its present form through ICH as the Medical Directory for Regulatory Activities (MedDRA).[22] These activities have become the domain of Case Management.

Next, moving to "output" in Figure 1, querying the database in the early years to answer internal (company) and external (generally regulatory) safety questions was an ad hoc, inconsistent, and highly variable process that was limited to the available functionality of the early databases. By the early 2000s, commercial database vendors had expanded the menu of capabilities significantly, and MedDRA had instituted Standardized MedDRA Queries[23] to facilitate different types of increasingly sophisticated database searches. These activities have become the domain of Signal Management.

The last component of this system is the Benefit–Risk Management. This component is an overall process that has been developed to account for all of a product's risk-related knowledge that has been organized and assessed as part of the case management and signal management activities and places it in the context of the product's benefit.[24]

AN OVERVIEW OF PV ACTIVITIES

Although a systems approach illustrates the relationships of the core disciplines, it does not show the actual breadth of scope of PV-related activities. Table 1 presents a discussion of these activities.

Protecting Human Subjects During the Conduct of Clinical Trials

One of the most important of the PV responsibilities is assuring the protection of human subjects participating in clinical trials (Table 1). The history of egregious failures in this regard means that all stakeholders must remain ever-vigilant. The human experimentation in World War II Germany and Japan and the Tuskegee syphilis "study" of 1932–1972 in the United States bear witness to these failings. Societal and institutional responses followed: the Nuremberg Code of 1947,[25] the Declaration of Helsinki of 1964,[26] and the Belmont Report of 1979.[27]

Stakeholders

In addition to PV, protecting human subjects is the business of the entire pharmaceutical research and development enterprise, each entire company and all

Table 1. Activities currently included in the scope of pharmacovigilance.

Category	Specific Activities/Functions	Phase(s)*
Supporting patient safety during the conduct of clinical trials	Informed consent, institutional review board, data monitoring committee	1–4
Selecting the first safe dose; first-in-human	Preclinical data, especially PK/PD parameters	1
Establishing the safety profile	Assessing all phases of development, focusing on dose-limiting toxicity, maximum tolerated dose, AEs of special interest, on-target and off-target toxicities	1–4
Communicating information to stakeholders	Maintaining standard formats: Investigator's Brochure, Company Core Data Sheet, package insert, patient package insert, ClinicalTrials.gov	1–4
Attending to surveillance activities	Determining relationships between drugs and adverse events through passive and active methods	1–4
Monitoring safety-related issues that involve the quality of the manufactured product	Conducting health hazard assessments for manufacturing deviations, complaints	1–4
Managing risk: REMS, RMP	Understanding benefit–risk across patient populations and uses	1–4
Maintaining inspection readiness	Preparation for scheduled and unscheduled inspections of department activities	1–4
Training	Clinical investigators; internal customers throughout the company; vendors	1–4
Advertising and promotion review	Assuring consistency with important safety information	4
Providing medical information to health care professionals	Support for professional queries regarding product complaints, AE reports, product use	4
Conducting due diligence	Understanding critical safety information about products being considered for merger, acquisition, or licensing activities	1–4

AE = adverse event; PK/PD = pharmacokinetics/pharmacodynamics; REMS = Risk Evaluation and Mitigation Strategy; RMP = Risk Management Plan.
*The phase(s) of the drug development process that include the described activities.

stakeholders: the clinical investigators, the journals that publish findings, the individual patients and their spokespersons and guardians, the regulators, and the research institutions that support and contribute to the work.

Concerns

In general, all stakeholders share, from different perspectives, concerns about patient safety. They also share concerns regarding patient privacy, accountability, communication, data integrity, study integrity, transparency, and information sharing.

Controls

Historically, disappointments, ethical failings, and tragedies have frequently led to statutory remedies, followed by promulgation of detailed regulatory processes and development of explanatory guidances for various items. These items include informed consent, institutional review boards in the United States and ethics committees in most other countries,

publication standards, and data monitoring committees.[28]

Role of PV

There is nearly global consistency of stakeholder participants, stakeholder concerns, and regulatory and institutional controls. PV professionals contribute review expertise to the key control documents concerning participants in clinical trials: informed consent forms for clinical studies, institutional review board documentation, and data monitoring committee deliberations. Related documents include the collection and reporting of safety information in the clinical study protocols. Much of this information has been standardized, such that templated materials are routinely used. However, periodic revisions are required, and PV professionals should be on the distribution list for participation in these updates.

Selecting the First Safe Dose (First-in-humans)

Crossing over from preclinical, animal studies to first-in-human studies, identified as T1 translational medical research,[29] is a significant step in the research and development enterprise; however, as the first segment of clinical investigations, it is also where ~90% of NMEs fail to obtain eventual regulatory approval.[30] Identifying the first safe dose to be administered to human subjects is a complicated process that reflects what has been learned about the pharmacokinetic and pharmacodynamic variables in the animal species that had been exposed to the new chemical entity (Table 1). It remains a step with significant risk.[31-34]

Role of PV

PV professionals are infrequently engaged directly in first-in-human studies and such studies are generally not considered a part of their professional remit. However, as these experiences show, they may be in an ideal position to play a critical role in seeing the whole picture and keeping all of the key stakeholders informed about developments.

Establishing the Safety Profile

In practice, an understanding of a new molecular entity's safety profile begins during animal studies (Table 1). The pharmacokinetic/pharmacodynamic studies provide key insights into the first major group of adverse events expected to be seen during clinical development: the likely on-target adverse events that are usually seen in a dose-dependent fashion during dose-escalation studies. These are the best predictors of human responses, and they direct the clinical and PV professionals to the types of physical examination parameters, laboratory tests, and investigations that should be followed during human studies. These adverse events eventually constitute ~80% of the adverse events that are described in the Investigator's Brochure and the prescribing information (PI), should the NME be eventually approved. An example is the predictable hypotension that is seen when a volunteer is dosed with increasing amounts of an antihypertensive drug.

The second major group of adverse events seen during development is the idiosyncratic, off-target event: it is generally uncommon, may be mild to severe, is rarely seen during animal studies, and the mechanism is usually unknown.

CIOMS has characterized these idiosyncratic events as falling into 1 of 2 groups[13]: (1) designated medical events; and (2) targeted medical events. Designated medical events are rare, serious, and most often attributable to a pharmacologic agent, although from across any of multiple distinct classes. Examples include drug-induced liver injury, hypoplastic/aplastic anemia, Stevens-Johnson syndrome, toxic epidermal necrolysis, torsade de pointes, and the hypersensitivity reactions of intravenously infused biologic agents. The second group of events, targeted medical events, is similarly rare and serious, although most often these events are attributable to a particular pharmacologic drug or class of drugs. Examples include the dry cough of angiotensin-converting enzyme inhibitors,[35] the tendon rupture of quinolone antibiotics,[36] and the renal or biliary calculi composed purely of atazanavir.[37]

After approval, several other types of events may be seen: (1) events related to chronic use beyond the durations studied during development (eg, the renal toxicity associated with chronic use of phenacetin[38]); (2) events that are delayed far beyond the periods of use (eg, the teratogenicity attributable to thalidomide taken early in pregnancy;[39] the clear cell carcinoma of the reproductive tract of women attributable to diethylstilbestrol seen in postmenarcheal female offspring of mothers who were prescribed the drug during their pregnancies decades earlier[40]); (3) withdrawal effects seen with narcotic drugs (eg, the

withdrawal after chronic use of opioids); and (4) failure of pharmacologic effect, not to be confused with failure of therapeutic effect (eg, the lack of any pharmacologic effect seen with counterfeit agents that consist of inert materials and with products diverted from proper commercial channels that may have become inactivated through improper handling or storage).

Role of PV

There is generally global consistency in collecting the information needed to establish and maintain safety profiles of drugs throughout a product's life-cycle. However, there are significant differences in perspectives in communicating information to stakeholders (discussed in the section entitled "Communicating Information to Stakeholders") and in maintaining product quality (discussed in the section entitled "Monitoring Safety-Related Issues of the Manufactured Product"). These uncommon categories of idiosyncratic adverse events are largely restricted to the vagaries of postmarketing experiences, and they are reminders of the general observation that drug-related safety concerns can arise from anywhere and at any time. PV professionals must maintain a "risk-awareness" of the class effects of both on-target and off-target effects of their products, beginning while NMEs are under development and extending throughout their life-cycles until eventual withdrawal from the market as obsolete pharmaceutical agents.

Communicating Information to Stakeholders

Clinical investigators, health care providers with prescribing privileges (including primarily physicians, nurse practitioners, and physician assistants), pharmacists, and patients represent the key stakeholders who have an interest in the information prepared by the company and the regulatory agency, based on the information submitted with the dossier (Table 1); for example, the New Drug Application of the US FDA or the Marketing Authorization Application of the European Union's European Medicines Agency (EMA). Each of the following documents addresses the needs of a particular stakeholder.

Investigator's Brochure

During clinical development, the Investigator's Brochure plays a critical role in informing all clinical investigators conducting studies on behalf of a pharmaceutical company of the current status of knowledge about not only the product under study but also, uniquely among these communication documents, the relevant drugs in class that may be harbingers of events to come for the product under study.[41] The Investigator's Brochure further provides a list of expected adverse events prepared by the company after careful review and assessment of reported events. The Investigator's Brochure is part of the ICH paradigm of guidances that the FDA and EMA have integrated into their regulatory documentation.

Company Core Data Sheet

For medicinal products, the content of the Company Core Data Sheet is prepared by the company, following regulatory templates; it reflects the company's fundamental understanding of the product regarding indications, dosing, pharmacology, safety profile, and other relevant information.[42] Alone among communication documents, this content is not subject to revision by agency regulators.

US Prescribing Information

The US PI (USPI) underwent a major revision in 2006, the Physician Labeling Rule.[43] It most notably includes a highlights section, which is unique among national prescription drug labels; additional changes include a table of contents and reordering of sections for usability and readability. It is the basis of information for health care professionals regarding the safe and effective use of the medicine under consideration. It is the basis for the Patient Package Insert (discussed below). It is also the key reference document for a company in preparation of advertising and promotional materials. The USPI is a carefully negotiated document between the company and the regulatory agency.

EU Summary of Product Characteristics

The EU Summary of Product Characteristics (EU SmPC) is the product label for prescription drugs in countries in the European Union.[44] Like the USPI, it is the basis of information for health care professionals regarding the safe and effective use of the medicine under consideration. It is also the basis for the Patient Information Leaflet (discussed below). It is similarly

the key reference document for preparation of advertising and promotional materials.

US Patient Package Insert, European Union Patient Information Leaflet

These information documents are hard copy handouts that come with many prescription medications and are based on the USPI/EU SmPC.[45,46] They are written in appropriate, nontechnical language. Their purpose is to help patients use the product safely and effectively and to minimize the risk of experiencing adverse events.

ClinicalTrials.gov

The FDA Modernization Act of 1997, section 801, established a registry (clinicaltrials.gov) of clinical trials information for federally and privately funded studies conducted under investigational new drug applications intended to evaluate the effectiveness for serious or life-threatening diseases or conditions.[47] The FDA Amendments Act of 2007 expanded this registry to include trial results. All information is provided by the party responsible for the clinical trial.

EMA Policy on Publishing Clinical Data

In October 2016, the EMA began to proactively publish clinical trial data submitted by pharmaceutical companies in support of their regulatory applications, once the evaluation process is completed.[48]

Role of PV

The process-related procedures for these documents are generally, although not always, managed by the regulatory affairs department of a company. However, responsibility for content falls to the respective departments that oversee preclinical animal studies, clinical indications, adverse events, product, and packaging. PV takes the lead in proposing changes to the safety profile of the product and must have robust governance structures and processes in place to support such proposals in content, form, and location in the respective user documents. It is also critical to appreciate that the labeled information sets the boundaries for advertising and promotion (discussed in the section entitled "Advertising and Promotion").

Attending to Surveillance Activities

Surveillance activities concern marketed products and directly include 2 of the core PV activities, case management and signal management (described earlier), which together inform the third, benefit–risk management. Signal management activities may be considered from the perspective of the information source (Table 1). For example: (1) passive receipt of adverse event reports from health care providers and consumers; (2) weekly surveillance of the world's literature for investigator-initiated study results and case reports; (3) data-mining methods for large, accessible drug safety databases; (4) manufacturing product complaints that may be associated with adverse events that could signal compromised product (counterfeit or diverted from approved channels); (5) patient registries required by regulatory authorities; and (6) regulatory agency–directed postapproval safety studies.

To complement the range of company-related signal management activities, the FDA established the Sentinel Initiative as part of the FDA Amendments Act of 2007. The FDA has rapid and secure access to large caches of electronic health care data, such as electronic health records, insurance claims data, and registries, as a source of information for signal detection activities. Presently, "[t]he FDA continues to develop Sentinel in stages to ensure the development of a long-term, sustainable system."[49]

In general, each source of potential safety issues requires a predefined process for frequency of investigation, assessment, potential follow-up, communication, and documentation. The outcome status of the process may be structured according to 1 of the following 3 courses: (1) file, with no further follow-up indicated; (2) revisit, determined to be at a predetermined interval (eg, 3 months for products early in development, 6 months for products recently marketed, and 1 year for products that have $\geq$5 years of marketed experience); or (3) determination of a safety signal that requires communication to stakeholders (eg, data monitoring committee members, clinical investigators, institutional review board [ethics committee] members, regulators) and updates of key documents (eg, clinical protocols, informed consent documents, Investigator Brochures, risk management plans, REMS documents, labeling documents).

Role of PV

This is a critical and highly dynamic set of activities for PV professionals that can require significant effort to assure timely execution. It is strongly identified with PV and leads to a great deal of interaction with

other company functions, including translational research, clinical trials, regulatory affairs, epidemiology, medical affairs, and manufacturing.

Monitoring Safety-related Issues of the Manufactured Product

PV may be called upon by Manufacturing to address issues that arise in any of the 3 segments of the manufacturing process: upstream sourcing of materials, during the manufacturing process itself, and downstream processes (Table 1). A brief description follows; a detailed discussion of this topic has been published elsewhere.[50]

Upstream

The heparin tragedy of 2008 has been the most serious incident involving manufacturing source material.[51] It began as a cluster of hypersensitivity reactions that resulted in >80 deaths among patients in the United States and Europe receiving heparin from what was eventually proven to be a single generic manufacturing source. Careful epidemiologic investigation of the clusters involving novel biochemical tools eventually identified the responsible chemical and tracked it back to financially motivated adulteration of porcine intestines from family workshops in China. Consequently, comparable legislation passed in the United States and the European Union now requires that each manufacturer maintain a facility identifier system that tracks the global supply chain to assure a documented pedigree of safe source materials.[52,53]

Manufacturing Process

The manufacturing process is subject to a range of possible deviations that could affect the long-term stability of the product that may not be detected until well after the product is released into distribution channels. PV professionals may be asked to provide a Health Hazard Assessment to determine the potential impact on patients.

Downstream

A significant regulatory gap exists between the highly regulated manufacturing process up until the release of finished product and the administration of finished product to patients who subsequently experience an adverse event, the process of which is also highly regulated. That gap includes transportation-related activities of product, including shipping, warehousing, and distribution to hospitals, pharmacies, clinics, and physician offices. The following can occur in this gap: undocumented temperature excursions for temperature-sensitive products that might compromise product integrity, damage to product packaging, and lack of chain of control that can be breached by malfeasant activities, including diversion of bona fide product and introduction of counterfeit product. The Drug Supply Chain Security Act of 2013 addresses many of these deficiencies.[52]

Role of PV

PV professionals must have a high index of suspicion when adverse event reports suggest compromised finished product, such as reports of damaged packaging, expired product, or lack of effect, or when these reports identify adverse events not consistent with the known safety profile of the product. Follow-up activities should be consonant with the degree of concern generated by the information.

Managing Risk

REMS: The Case-by-case Approach

With the Prescription Drug User Fee Act III of 2002, the FDA agreed to provide guidance to industry on risk management activities; this resulted in the Risk Minimization Action Plan. With the FDA Amendments Act of 2007, the FDA was granted the specific authority to require a company with a newly approved or previously approved drug (or biologic) that has a significant adverse event to develop a risk management plan that uses risk minimization strategies beyond professional labeling to provide assurance that the drug's benefits outweigh its risks.[54,55] Mechanisms may include prescriber training, pharmacist certification, restricted dispensing, patient testing, and patient registries (Table 1). This approach has experienced growing pains. For example, the Department of Health and Human Services Office of Inspector General reported in 2013 that only 7 (14%) of 49 REMS programs met all goals and that the "FDA had not identified reliable methods to assess the effectiveness of REMS."[56]

Risk Management Plan: The Comprehensive Approach

The EMA notes that at the time of authorization of a new drug or biologic agent, the benefit–risk balance

is demonstrably favorable; however, the restricted experience in numbers of patients and duration of use limit the understanding of the safety profile. Thus, the risk management plan contains the following: (1) the safety specification (important identified risks, important potential risks, and missing information); (2) the PV plan (activities to quantify clinically relevant risks and to identify new adverse reactions); and (3) the risk minimization plan (implementation of risk minimization measures).[57] Also of importance, postauthorization safety studies and postauthorization efficacy studies may be required under certain conditions.[15] As with REMS, the risk management plan has experienced growing pains, as demonstrated by industry comments that accompanied the revisions for Module V—Risk Management Systems (Revision 2).[15,58]

Role of PV

The divergent regulatory approaches between the FDA and the EMA require PV professionals to be particularly alert and attentive to the occurrence of each new signal, as it may have different implications for how it is to be managed in each regulatory jurisdiction. Lis et al[59] have provided a comparison of approaches.

Maintaining Inspection Readiness

Regulatory authorities are obligated to assure stakeholders and the public that there is compliance with regulations by companies that provide products for markets for which they have oversight. This goal is accomplished through inspections (Table 1). For pharmaceuticals, the different national and regional jurisdictions of regulatory authorities around the world have much in common in their inspection practices. In the most general sense, there are 2 fundamental questions: (1) Are there adequate processes in place to assure compliance with the regulations? and (2) Is the company in compliance with its own processes? To meet these expectations, it is essential to be inspection-ready, not as an individual event of preparation at the time of an impending inspection but as a corporate cultural process of readiness maturation that is undertaken every day.[60,61]

There are 3 general types of inspections: (1) preapproval at the time that a company submits a new marketing application; (2) a routine inspection that is risk-based in frequency and usually conducted at least every 2 years; and (3) a "for-cause" inspection to evaluate specific problems that have come to an agency's attention. FDA follows its Guide to Inspections of Quality Systems.[62] This guide includes 4 major elements: (1) management controls; (2) design controls; (3) corrective and preventative actions; and (4) product and process controls. The EMA follows its Module III—Pharmacovigilance Inspections, located in the Guideline on Good Pharmacovigilance Practices.[63]

Training

Regulatory audits increasingly stress the preventative value of training in pharmaceutical companies: varying degrees of thoroughness and frequency depend upon the regulatory expectations and depth in the organization, including in its vendors and contractors (Table 1). Regulators hold the entire company and its vendors responsible for reporting adverse events. The PV department commonly prepares, approves, and maintains the training and assessment materials.

Advertising and Promotion Review

The approved prescribing information, such as the USPI or the EU SmPC, serves as the basis for advertising and promotional activities that a company undertakes (Table 1). The FDA and EMA have comparable approaches to advertising and promotional activities.[64,65] Most mid-to-large pharmaceutical companies have institutional governance processes in place to review and approve the use and dissemination of advertising and promotional materials. These committees/boards are usually composed of a lawyer, a regulatory affairs professional, a marketer, a clinician, and only uncommonly a PV physician. However, it is the PV physician who is closest to the safety profile of the product under discussion and who can best assess the fairness of the benefit—risk balance being presented in the advertising and promotional materials. The company department responsible for the advertising and promotional activities may request safety information to support a competitive advantage, which is a superiority claim, in a particular advertising campaign. Regulators generally require specifically designed clinical trials to support the use of safety information in such activities.

Direct-to-Consumer Advertising

Beginning in 1985, the FDA gradually permitted direct-to-consumer (DTC) advertising.[66,67] New Zealand is the only other country to permit DTC advertising.[68] PV departments usually play no part in DTC activities.

Medical Information

Medical information has increasingly become a subspecialty of its own, with diminishing contributions from PV (Table 1). Furthermore, it has commonly come under the aegis of Medical Affairs, which generally provides the medical expertise for responses to health care provider queries. The role of PV is generally limited to consultation at the request of medical information about infrequently asked marketed product safety issues that arise from health care providers who contact the company. In this case, carefully balanced statements are required, keeping in mind that responses should not represent "new information" that would otherwise be required to go through the company's governance processes for review, approval, and submission to regulatory authorities for labeling consideration.

Due Diligence

PV professionals have become increasingly important contributors to due diligence activities in the mergers and acquisition and in-licensing spaces (Table 1). This scenario is especially the case for large capitalized pharmaceutical and biotech companies for which basic science has been playing a shrinking strategic role.

For products in development, items that require PV review usually include the following: officially signed reports of all nonclinical animal studies and all human studies; a summary description of all adverse events, especially serious adverse events, in the company's database; a summary of all periodic reports of safety issues and risk management activities; a summary of all PV operational processes; product labels of all competitive marketed products in class; all regulatory correspondence and all internal, external, and regulatory inspection reports of the PV department and its database(s); and a relatively exact distribution of the numbers of subjects/patients who have received the product during development.

For products on the market, the review should additionally include a summary of all manufacturing product complaints, as well as a more general estimate of the numbers of patients who have been prescribed the product. If a site visit is undertaken, interviews of the PV management and supervisory professionals should also be conducted.

During the process of assessing the available information, the ultimate goal is to characterize the risks associated with the product in its competitive environmental context: what is known about the product and what can be expected to be learned given its place in a product's life-cycle, what is known about the other products in its class, what is not known but can be learned or otherwise acquired, the quality of the information, and the institutional and regulatory gaps in its processes.

DISCUSSION

This overview reflects the author's 3+ decades of career experience as a regulator and member of the Senior Executive Service in the FDA, as a manager and executive in the pharmaceutical industry (including PV), and as a faculty member in a US medical school. From this perspective, there have been tremendous evolutionary, some might even say "revolutionary," changes in the scope and depth of PV activities over the past several decades. Core activities have been established: case management, signal management, and benefit–risk management. Individual case review activities that began in postmarketing have expanded to include aggregate review and analytics, and have extended to involve products in development and, in some companies, into preclinical studies, particularly as information becomes available about the safety profile in animals. The goal is to develop an understanding about how that information can inform clinical investigators regarding the expected adverse events in the first-in-human studies.

The concept of risk and its related concept, benefit–risk, has been introduced into the evaluation paradigm. Risk itself has evolved into a full-fledged paradigm of its own. The key elements involve risk identification, risk analysis, risk mitigation, and risk communication. For PV, this paradigm could also be broadened to include risk awareness at the front end, with the intent of developing an explicit recognition

and understanding of all the therapeutic agents likely to be available in the treatment of the disease or stage of disease under consideration. In this way, potential drug-drug interactions become readily apparent, especially as new drugs-in-class and new classes of drugs continue to come to the market over time, and benefit–risk assessments are more likely to include relevant treatments.

The range of risk-related mitigation and preventative activities includes the following: (1) management of adverse events identified during development or collected in postmarketing that become included in the label; (2) vetting of advertising and promotional offerings for the products that depend on the label for its promotional statements; and (3) most recently, analysis of manufacturing complaints that may be associated with clusters of reports of adverse events, either because of product in bona fide distribution channels that unknowingly becomes compromised during transportation or storage, product that is diverted and becomes compromised, or product that is counterfeit material in origin.

Mammi et al[69] provide an overview of PV in pharmaceutical companies in Europe, using the EMA regulatory framework as the basis for discussing the expected responsibilities and activities.

Speculation regarding future trends within PV must begin with biomedical informatics, as has been discussed elsewhere.[70] Biomedical informatics will accelerate efficiencies through the following: (1) greater speed; (2) cheaper, denser, and more massive storage; and (3) more sophisticated software, especially through artificial intelligence and machine learning. Biomedical informatics will facilitate the development of—and in turn, perform—more extensive analytics. There is the potential for adverse reactions to be identified sooner and labeled earlier in the product's life-cycle. Mitigation-driven intervention will move closer to prevention. Artificial intelligence and machine learning will continue to expand these trends.[71,72]

All of the rapidly evolving technology will itself drive faster and broader acquisition by PV professionals of diverse quantitative skill sets: probability, biostatistics, epidemiology, statistical process control (control charting), quality management (root cause analysis, and failure mode effects analysis), computer science, and analytics. Other disciplines will find their way into the PV arena: behavioral economics, psychology, communication, and project management. Whatever skills that PV professionals do not have will be learned on-the-job or acquired through ad hoc courses or organized credentialing programs.

PV itself can be expected to continue to evolve. Tracking of everything will improve: wearable technologies can be expected to be tapped to support products in development; surveillance of social media will be routinized. Increasingly massive and disparate databases will be routinely linked. PV will make inroads in mitigation of the consequences of errors and possibly prevention of errors, for example, by requiring labels of high-risk products to include information on dialyzability. Greater attention may be brought to the proper disposition of drug products after use to reduce inadvertent exposure at the individual and population level to biologically active products, antibiotic products, abuse potential drugs, and other drugs harmful to humans or the environment. There will be stakeholder pressure to improve evaluative methods to answer the essential question: what is the evidence for what these activities are accomplishing?

CONCLUSIONS

This overview shows how the discipline of PV has undergone dramatic changes since the thalidomide tragedy and aftermath of the late 1950s and early 1960s. Every aspect of PV has been affected, beginning with the establishment of 3 core activities: case management for the standardization of adverse event information, regardless of source, that is entered into highly sophisticated databases; signal management for querying these databases to detect drug-event relationships and to manage their ongoing activities; and benefit–risk management to implement processes that mitigate patient risk to maintain a favorable benefit–risk balance.

Concurrently, PV has built out multiple other activities, often in collaboration with other departments. Some examples include the following: supporting patient safety during the conduct of clinical trials through well-developed patient consent forms and institutional review board documents, with Clinical Development; selecting the first safe dose for use in humans through analysis of nonclinical pharmacokinetic/pharmacodynamic studies, with Clinical Pharmacology; developing the range of documents that describe the safety profile of the

experimental agent for use by an array of stakeholders, with Regulatory Affairs; and monitoring the safety-related issues of the manufactured product, such as counterfeit and diverted product, with Manufacturing.

Finally, we should recognize that nongovernmental organizations such as CIOMS and ICH have globally played key roles in building out and institutionalizing the infrastructure of concepts, definitions, and processes for regulatory, clinical, PV, and manufacturing use. Thus, what have been generally unconnected or unrelated disciplines are finding their way into mainstream PV activities interfacing with other disciplines. The future of PV is bright.

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CONFLICTS OF INTEREST

Versions of this content have been presented previously: Tufts Center for the Study of Drug Development Post-Graduate Course, February 2014 to 2018; Marcus Evans, September 17, 2017, Westlake Village, California. These presentations were made while Dr. Beninger was employed by Genzyme, a Sanofi company, which provided salary, bonus, and stock options during his employment, until his retirement in April 2017. The author has otherwise received no financial support from industry; the sponsoring company (Marcus Evans) provided travel and accommodations for the September 17, 2017, meeting. No one else influenced choice of content during the preparation of the manuscript. The author has indicated that he has no other conflicts of interest regarding the content of this article.

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